

Extended Kalman Filter for Orbital State Estimation

Description

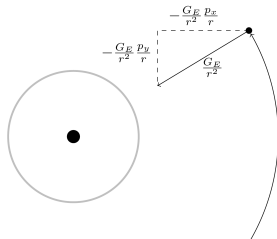
This project implements an Extended Kalman Filter for estimating the state of a satellite in 2D orbit about the Earth. The system dynamics are nonlinear due to gravitational acceleration, and the measurement model is nonlinear because the sensor provides only the bearing angle to the satellite. The filter estimates position and velocity over an eight-hour orbit using recursive linearization of both the dynamics and measurement models.

Problem Formulation

A satellite orbits the Earth. For simplicity, we will say that the gravitational acceleration on the satellite is G_E/r^2 , where we use the value $G_E = 3.986 \times 10^{14} m^3/s^2$ and r is the total distance from earth to the satellite, $r = \sqrt{p_x^2 + p_y^2}$. The earth is at position (0,0) and at this origin is a sensor that only measures the bearing to the satellite. Our state vector is the x and y position and velocity values:

$$\underline{x} = [p_x \quad p_y \quad v_x \quad v_y]^T$$

The acceleration of G_E/r^2 is just the magnitude. The vector of acceleration is the magnitude multiplied by the negative unit vector of position: p_x/r and p_y/r :



So, the dynamics equations are:

$$\begin{aligned} \dot{p}_x &= v_x + \nu_{p_x} \\ \dot{p}_y &= v_y + \nu_{p_y} \\ \dot{v}_x &= a_x + \nu_{v_x} \\ \dot{v}_y &= a_y + \nu_{v_y} \end{aligned}$$

where a_x and a_y are defined as

$$\begin{aligned} a_x &= -\frac{G_E p_x}{(p_x^2 + p_y^2)^{3/2}} \\ a_y &= -\frac{G_E p_y}{(p_x^2 + p_y^2)^{3/2}} \end{aligned}$$

and where the $\nu \sim N(0, \mathbf{Q})$ values are noise. In the discrete domain, using forward Euler integration, the dynamics equations are:

$$\begin{aligned} p_{x,k+1} &= p_{x,k} + v_{x,k}T + \nu_{p_x}\sqrt{T} \\ p_{y,k+1} &= p_{y,k} + v_{y,k}T + \nu_{p_y}\sqrt{T} \\ v_{x,k+1} &= v_{x,k} + a_{x,k}T + \nu_{v_x}\sqrt{T} \\ v_{y,k+1} &= v_{y,k} + a_{y,k}T + \nu_{v_y}\sqrt{T} \end{aligned}$$

The measurement equation is:

$$z = \text{atan2}(p_x, p_y) + \eta$$

where $\eta \sim N(0, R)$.

Extended Kalman Filter

Because both the dynamics and measurement models are nonlinear, state estimation is performed with an Extended Kalman Filter. At each time step, the current estimate is propagated through the nonlinear dynamics, and the covariance is propagated using the Jacobian of the dynamics.

The measurement update uses the nonlinear bearing model together with the Jacobian of the measurement function. The innovation is computed as the difference between the actual and predicted bearing measurements. Since bearing is an angle, the innovation must be wrapped to the interval $[-\pi, \pi)$ before applying the update.

Dynamics Jacobian

The nonlinear dynamics are

$$\phi(x) = \begin{bmatrix} p_x + v_x T \\ p_y + v_y T \\ v_x + a_x T \\ v_y + a_y T \end{bmatrix}$$

The Jacobian of the dynamics is

$$\Phi = \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ G_E T \frac{2p_x^2 - p_y^2}{(p_x^2 + p_y^2)^{5/2}} & \frac{3G_E T p_x p_y}{(p_x^2 + p_y^2)^{5/2}} & 1 & 0 \\ \frac{3G_E T p_x p_y}{(p_x^2 + p_y^2)^{5/2}} & G_E T \frac{2p_y^2 - p_x^2}{(p_x^2 + p_y^2)^{5/2}} & 0 & 1 \end{bmatrix}$$

Measurement Jacobian

The measurement model is

$$h(x) = \text{atan2}(p_y, p_x)$$

The Jacobian of the measurement model is

$$H = \begin{bmatrix} \frac{-p_y}{p_x^2 + p_y^2} & \frac{p_x}{p_x^2 + p_y^2} & 0 & 0 \end{bmatrix}$$

Results

Figure 1 shows the true orbital path together with the estimated path from the Extended Kalman Filter.

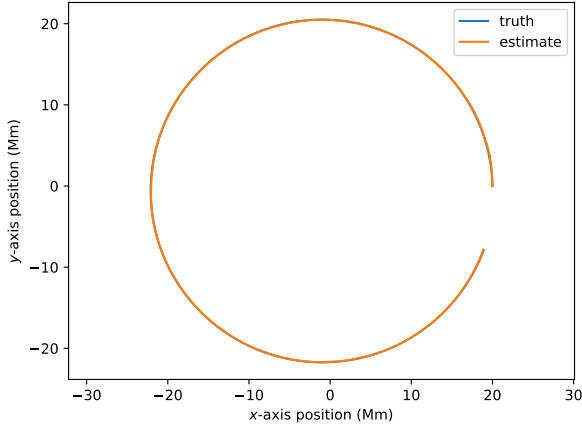


Figure 1: True and estimated orbital paths

Figures 2 and 3 show the position and velocity estimation errors over time along with their corresponding $3-\sigma$ bounds from the filter covariance.

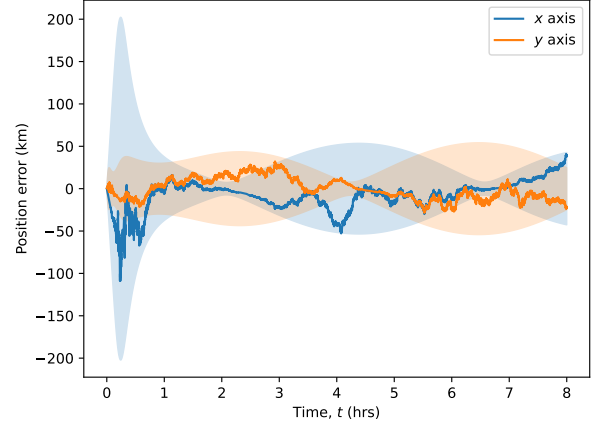


Figure 2: Position estimation errors

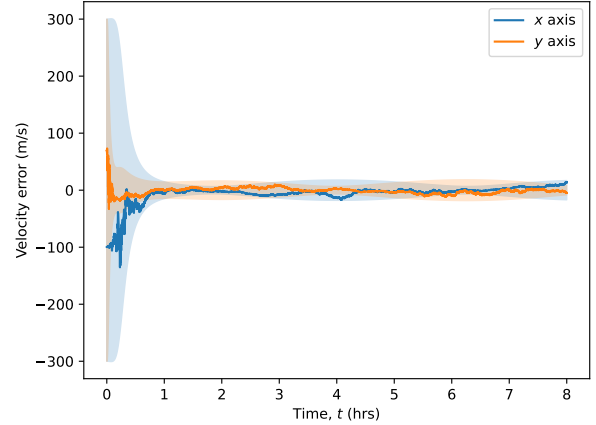


Figure 3: Velocity estimation errors

Discussion

The results show that the Extended Kalman Filter is able to track the satellite state despite nonlinear dynamics and a nonlinear bearing-only measurement model. The position and velocity errors remain bounded, and comparison with the $3-\sigma$ bounds provides insight into the consistency of the covariance propagation.

A key part of this implementation is the use of analytical Jacobians for both the dynamics and measurement models. Another important detail is angle

wrapping in the innovation, since unwrapped bearing residuals can create large artificial errors and destabilize the filter.

This project highlights the central EKF idea: a nonlinear estimation problem can be handled by propagating the full nonlinear state while locally linearizing the system for covariance calculations.